Chemistry and ice processes

Surface chemistry

Surface chemistry relies on many different parameters, with a complex interplay.

a. Calculate the diffusion prefactor ν of a hydrogen atom with $E_{\text{bind}} = 650$ K using the Hasegawa-Herbst equation:

$$\nu = \sqrt{\frac{2 n_{\text{sites}} k_{\text{B}} E_{\text{bind}}}{\pi^2 m}}.$$

Assume the site density on the grains is $n_{\text{sites}} = 1.5 \times 10^{15}$ cm^{-2}.

Hint: The Hasegawa-Herbst equation for prefactors gives values of 10^{12}–10^{13} s^{-1} *for most species.*

b. Calculate the diffusion rate constant k_{diff} of a hydrogen atom with $E_{\text{diff}} = 325$ K at $T = 10$ K (assuming diffusion is a completely thermal process). Use the prefactor from exercise a.

c. Calculate the tunneling probability of the hydrogen abstraction reaction $CH_3OH + H \longrightarrow CH_3O + H_2$ with a barrier $E_{\text{reac}} = 1000$ K. Assume the barrier width a is 1.4 Å. The tunneling probability can be calculated as

$$P_{\text{reac}} = \exp\left(-\frac{2a}{\hbar}\sqrt{2\mu k_{\text{B}} E_{\text{reac}}}\right)$$

d. Calculate the competition fraction f_{comp} between this reaction and the diffusion of the hydrogen atom. This can be calculated as

$$f_{\text{comp}} = \frac{\kappa_{\text{reac}}}{\kappa_{\text{reac}} + k_{\text{diff}}}$$

You can interpret f_{comp} as the probability that the reaction will occur before the hydrogen atom will move away.

In rate equation based models (such as astrochemical models), we only work with the concentrations (abundances) of species, and not with the individual atoms and molecules.

e. Consider that there is only one set of reactants. What is the probability that $CH_3OH + H$ will form $CH_3 + H_2O$ before it forms $CH_3O + H_2$? Assume that the probability of tunneling to form $CH_3 + H_2O$ is $P_{\text{reac}} = 10^{-9}$.

Hint: This can be calculated as the ratio of the average time $\tau = 1/k$ *it takes for each reaction.*

f. What does this mean for the "accretion limit", when there is only one set of reagents on the entire ice of a grain (or even less than one on average)? How do astrochemical models differ from a more realistic, particle-based (e.g. kinetic Monte Carlo) view in this situation?

Solution:

a. Filling in the numbers given, you get

$$\begin{aligned}\nu &= \sqrt{\frac{2 \cdot 1.5 \times 10^{19}\ \mathrm{m}^{-2} \cdot 1.380649 \times 10^{-23}\ \mathrm{m}^2\ \mathrm{kg\ s}^{-2}\ \mathrm{K}^{-1} \cdot 650\ \mathrm{K}}{\pi^2 \cdot 1.6735575 \times 10^{-27}\ \mathrm{kg}}} \\ &= 4.0 \times 10^{12}\ \mathrm{s}^{-1}.\end{aligned}$$

b. Diffusion is a completely thermal process, so the rate constant can be described by an Arrhenius equation,

$$k_{\mathrm{diff}} = \nu \cdot \mathrm{e}^{-E_{\mathrm{diff}}/T}.$$

Filling in the values, along with the prefactor we got from a, we get $k_{\mathrm{diff}} = 0.031\ \mathrm{s}^{-1}$.

c. The reduced mass of the reagents is

$$\mu = \frac{m_1 m_2}{m_1 + m_2} = \frac{1 \cdot 32}{1 + 32} = 0.97\ \mathrm{amu} = 1.61 \times 10^{-27}\ \mathrm{kg}$$

Filling in the values, we get

$$\begin{aligned}P_{\mathrm{reac}} &= \exp\left(-\frac{2 \cdot 1.4 \times 10^{-10}\ \mathrm{m}}{1.054 \times 10^{-34}\ \mathrm{kg\ m}^2\ \mathrm{s}^{-1}} \cdot \right. \\ &\left. \sqrt{2 \cdot 1.61 \times 10^{-27}\ \mathrm{kg} \cdot 1.380649 \times 10^{-23}\ \mathrm{m}^2\ \mathrm{kg\ s}^{-2}\ \mathrm{K}^{-1} \cdot 1000\ \mathrm{K}}\right) \\ &= \exp(-17.7) \\ &= 2.03 \times 10^{-8}.\end{aligned}$$

d. $\kappa_{\mathrm{reac}} = \nu \cdot P_{\mathrm{reac}} = 8.19 \times 10^4\ \mathrm{s}^{-1}$, such that

$$f_{\mathrm{comp}} = \frac{8.19 \times 10^4\ \mathrm{s}^{-1}}{8.19 \times 10^4\ \mathrm{s}^{-1} + 0.031\ \mathrm{s}^{-1}} \approx 1,$$

which means that the reaction is far more likely to occur than the hydrogen atom diffusing away again.

e. This ratio is:

$$\frac{\nu \cdot P_{\mathrm{CH_3OH + H \longrightarrow CH_3O + H_2}}}{\nu \cdot P_{\mathrm{CH_3OH + H \longrightarrow CH_3 + H_2O}}} = \frac{2.03 \times 10^{-8}}{1 \times 10^{-9}} = 20.3,$$

which means that it is about 20 times more likely to form CH_3O and H_2 than it is to form CH_3 and H_2O.

f. In the kinetic Monte Carlo framework, the reaction to form CH_3 and H_2O would likely not occur, so CH_3 would not be formed by this reaction. But, in the rate-equation based framework CH_3 is produced, just with a quantity of about 20 times lower.

Code for solution:

```python
import numpy as np
from scipy.constants import Boltzmann, angstrom, atomic_mass, centi, hbar, pi

if __name__ == "__main__":
    n_sites = 1.5e15  # cm-2
    E_bind = 650  # K
    nu_H = np.sqrt(
        (2 * n_sites / (centi**2) * Boltzmann * E_bind) / (pi**2 * 1 * atomic_mass)
    )
    print(f"a. Prefactor: {nu_H:.2e}")

    T = 10.0  # K
    E_diff = 325  # K
    diffusion_rate_constant = nu_H * np.exp(-E_diff / T)
    print(f"b. Diffusion rate constant k: {diffusion_rate_constant:.2e}")

    a = 1.4 * angstrom
    E_reac = 1000  # K
    reduced_mass = 32 * 1 / (32 + 1) * atomic_mass
    P_reac = np.exp(-2 * a / hbar * np.sqrt(2 * reduced_mass * Boltzmann * E_reac))
    print(f"c. Tunneling probability P_reac: {P_reac:.2e}")

    f_comp = (P_reac * nu_H) / (P_reac * nu_H + diffusion_rate_constant)
    print(f"d. Competition fraction f_comp: {f_comp:.4f}")

    rate_ratio = P_reac / 1e-9
    print(f"e. Probability of forming CH3 + H2O: {rate_ratio}")
```

Reaction rate analysis

Analyzing the formation and destruction reactions of a certain species at some set of physical conditions can help to give more information about their origin.

a. Analyze the main formation and destruction reactions of CH_3OH in the gas-phase at $T = 10$ K, $n_{\mathrm{H}} = 10^5$ cm^{-3}, $F_{\mathrm{UV}} = 1$ Habing and $\zeta = 1.3 \times 10^{-17}$ s^{-1}, after 10^5 years. Plot the 5 fastest formation and destruction reactions.

   Hint: The reaction rate constants k during a model run can be obtained by setting `return_dataframe` *and* `return_rates` *to* `True`*. See* https://uclchem.github.io/v4.0.0-rc1/notebooks/4_chemical_analysis.html*. Note that this returns the reaction* ***rate constants****, and not the reaction* ***rates****.*

b. What does this tell you about the gas-phase abundance of CH_3OH at these physical conditions? What is it mainly dictated by?

c. Create a plot of the main formation and destruction processes of HNC over time from 10^0 to 10^6 years. Take the top 5 formation and destruction routes at any time (i.e. highest maximum rate).

Solution:

a. See Fig. 1. From this figure, it is clear that the main formation route of CH_3OH is the desorption from the ices due to cosmic ray-induced UV. Gas-phase processes do contribute some, but not nearly as much. Thermal desorption is neglible (as expected from the high binding energy of $\#CH_3OH$ of 5000 K and the low temperature).

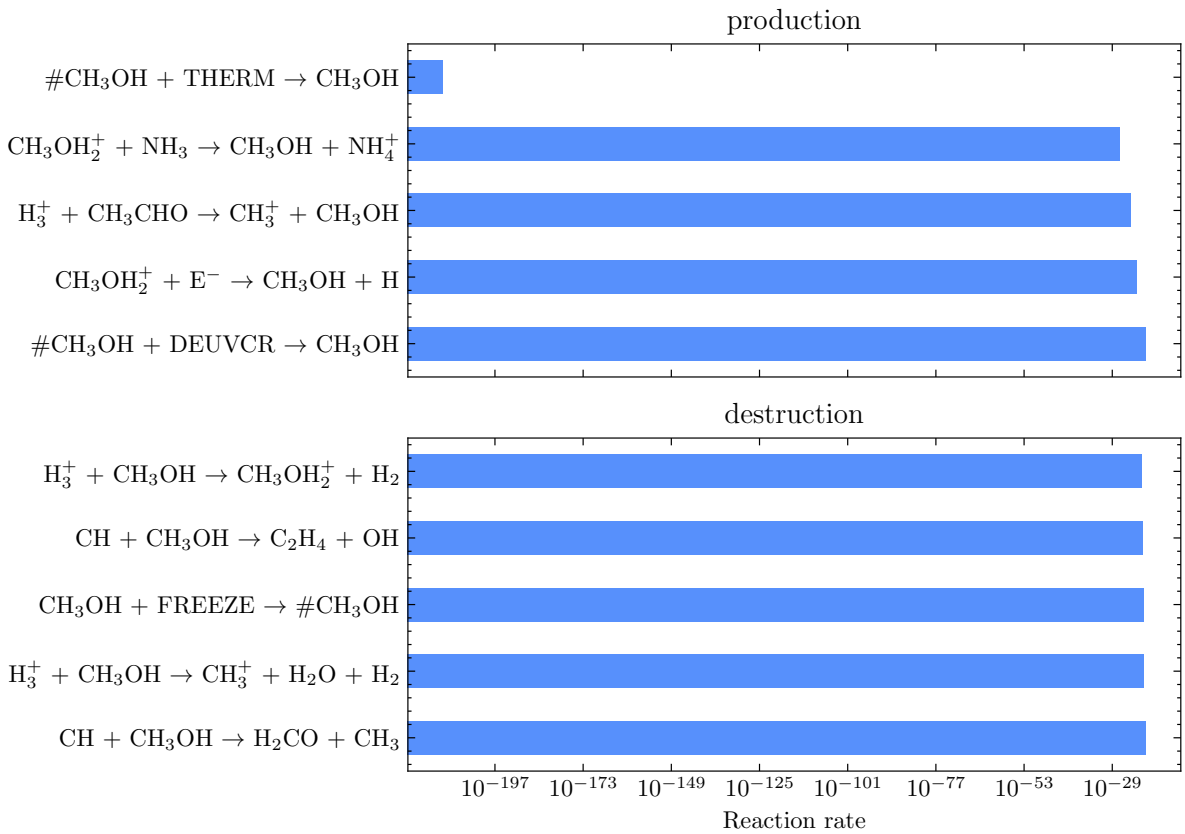

Figure 1: Main formation and destruction reactions of CH_3OH gas at $T = 10$ K, $n_H = 10^5$ cm^{-3}, $F_{UV} = 1$ Habing and $\zeta = 1.3 \times 10^{-17}$ s^{-1}, after 10^5 years.

b. It seems that the gas-phase abundance is mainly determined by its formation on the ice, and subsequent desorption due to cosmic ray-induced UV. CH_3OH is mostly destroyed in the gas-phase.

c. See Fig. 2.

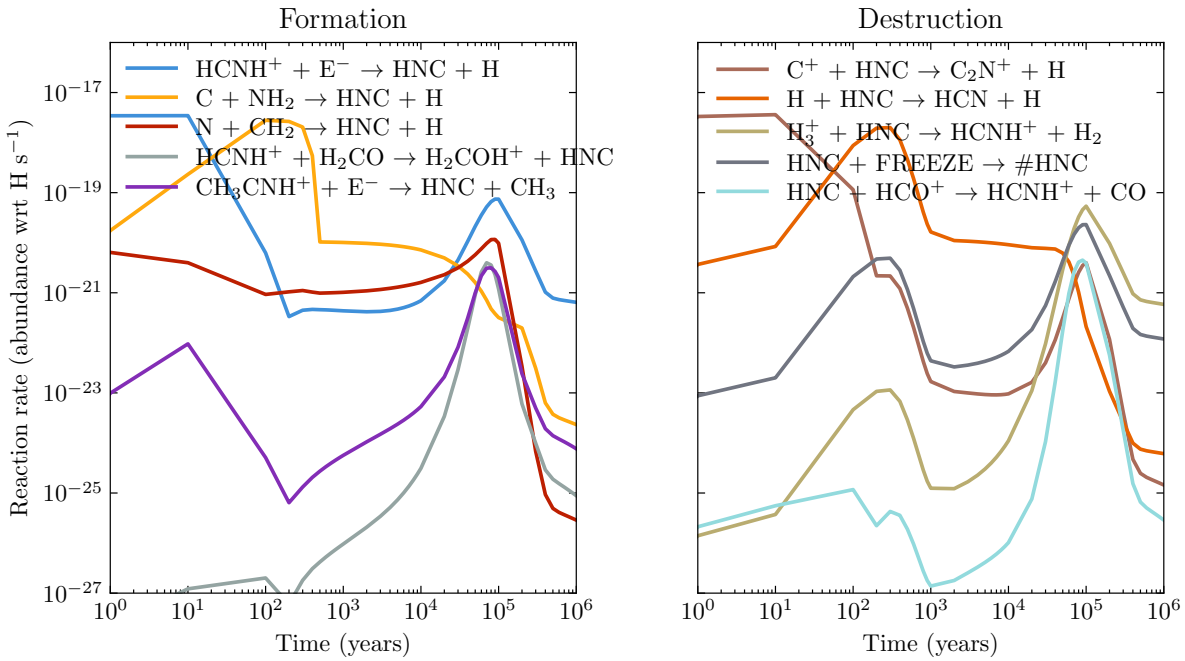

Figure 2: Main formation and destruction reactions of HNC gas at $T = 10$ K, $n_{\rm H} = 10^5$ cm^{-3}, $F_{\rm UV} = 1$ Habing and $\zeta = 1.3 \times 10^{-17}$ s^{-1} over time.

Code for solution:

```python
import numpy as np
import matplotlib.pyplot as plt
import uclchem
from uclchem.analysis import (
    get_production_and_destruction,
    rate_constants_to_dy_and_rates,
)
from uclchem.makerates.network import load_network_from_csv
from uclchem.plot import plot_rate_summary
from python_tools import convert_reaction_to_latex_string, get_colors


if __name__ == "__main__":
    plt.style.use(["science", "font_latex", "color_petroff_6"])

    # a.
    # set a parameter dictionary for model
    params = {
        "endAtFinalDensity": False,
        "freefall": False,
        "writeStep": 1,
        "initialDens": 1e5,
        "initialTemp": 10.0,
        "finalTime": 1e6,
        "zeta": 1,
        "radfield": 1,
    }

    physics_df, chemistry_df, rates_df, _, _, _ = uclchem.functional.cloud(
        param_dict=params, return_dataframe=True, return_rates=True
    )

    dy, rates_by_reaction_df = rate_constants_to_dy_and_rates(
        physics_df, chemistry_df, rates_df, network=load_network_from_csv()
    )

    spec = "CH3OH"
    production, destruction = get_production_and_destruction(spec, rates_by_reaction_df)

    time_index = np.argmin(np.abs(physics_df["Time"].to_numpy() - 1e5))

    plot_rate_summary(
        production,
        destruction,
        time_index,
        xlabel="Reaction rate (abundance wrt H s$^{-1}$)",
    )
    fig = plt.gcf()
    for axis in fig.axes:
        ticklabels: list[plt.Text] = axis.get_yaxis().get_ticklabels()
        axis.get_yaxis().set_ticklabels(
            [convert_reaction_to_latex_string(label.get_text()) for label in ticklabels]
        )
    plt.tight_layout()
```

```python
    plt.savefig("rates_CH3OH_gas.pdf")
    plt.show()

    # c.
    spec = "HNC"
    production, destruction = get_production_and_destruction(spec, rates_by_reaction_df)
    max_production_at_any_timestep = np.max(np.asarray(production), axis=0)
    max_production_reactions_overall = np.argsort(max_production_at_any_timestep)[::-1]
    production_reactions = production.columns

    max_destruction_at_any_timestep = np.max(np.asarray(destruction), axis=0)
    max_destruction_reactions_overall = np.argsort(max_destruction_at_any_timestep)[
        ::-1
    ]
    destruction_reactions = destruction.columns

    max_to_plot = 5

    plt.style.use(["color_petroff_10"])
    colors = get_colors()

    fig, axs = plt.subplots(1, 2, sharex=True, sharey=True, figsize=(7, 4))
    axs[0].set_title("Formation")
    for i in range(max_to_plot):
        reaction = production_reactions[max_production_reactions_overall[i]]
        axs[0].plot(
            physics_df["Time"],
            production[reaction],
            label=convert_reaction_to_latex_string(reaction),
            c=colors[i],
        )

    axs[1].set_title("Destruction")
    for i in range(max_to_plot):
        reaction = destruction_reactions[max_destruction_reactions_overall[i]]
        axs[1].plot(
            physics_df["Time"],
            destruction[reaction],
            label=convert_reaction_to_latex_string(reaction),
            c=colors[i + max_to_plot],
        )

    axs[0].set_xlim([1e0, 1e6])
    axs[0].set_ylim([1e-27, 1e-16])

    axs[0].set_xscale("log")
    axs[0].set_yscale("log")

    axs[0].set_xlabel("Time (years)")
    axs[1].set_xlabel("Time (years)")
    axs[0].set_ylabel("Reaction rate (abundance wrt H s$^{-1}$)")

    axs[0].legend()
    axs[1].legend()
```

```
    plt.tight_layout()
    plt.savefig(f"{spec}_rate_analysis.pdf")
    plt.show()
```

Sensitivity Analysis

Astrochemical models make many assumptions about the chemistry of ices. UCLCHEM allows you to test (some of) those assumptions relatively easily. Run all models here at the same physical conditions as in Reaction rate analysis.

a. What are the main destruction and formation reactions of CH_3OH surface ice at 10 K? Use a similar workflow as in Reaction rate analysis.

   Hint: The "surface" adsorbed species is indicated by prefixing a "#" in UCLCHEM.

b. Vary the diffusion to binding ratio χ between 0.25 and 0.75 in steps of 0.05. What effect does this have on the CH_3OH ice (surface and bulk combined) abundance? Create a plot with χ on the x-axis, and the CH_3OH abundance at 10^5 years on the y-axis.

   Hint: The diffusion to binding energy ratio corresponds to `diffToBindRatio` *in the parameter dictionary.*

c. Changing χ changes all of the diffusion barriers, which can make it hard to disentangle the effect of changing different diffusion barriers. Instead, vary the hydrogen diffusion barrier between 162.5 and 487.5 K in steps of 32.5 K. What effect does this have on the CH_3OH ice abundance? Create a similar plot as in b.

   Hint: You can change the H diffusion barrier using `HdiffusionBarrier` *in the parameter dictionary.*

d. Changing χ between 0.25 and 0.75 with steps of 0.05 and changing $E_{\text{diff,H}}$ between 162.5 and 487.5 K in steps of 32.5 K has the same effect on the hydrogen diffusion barrier (0.25×650 K $= 162.5$ K). Compare the plots you made in b and c (it is convenient to have the same y-axes in the two plots). What does this tell you about the main important chemical parameter for the abundance of $\#CH_3OH$ (at these physical conditions)?

Solution:

a. See Fig. 3. The main formation routes are reactions of a radical with atomic hydrogen, whereas the main destruction routes are

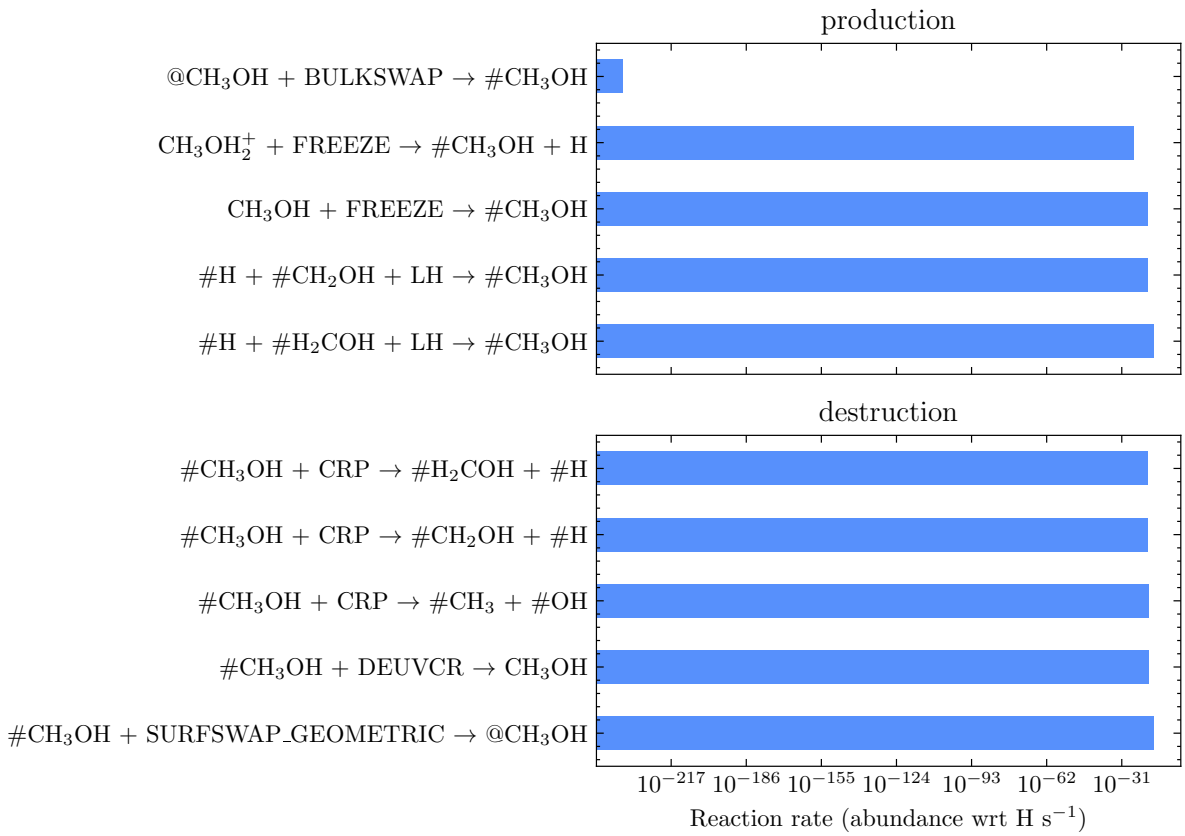

Figure 3: Main formation and destruction reactions of surface CH_3OH at $T = 10$ K, $n_H = 10^5$ cm^{-3}, $F_{UV} = 1$ Habing and $\zeta = 1.3 \times 10^{-17}$ s^{-1}, after 10^5 years.

b. See Fig. 4.

c. See Fig. 5.

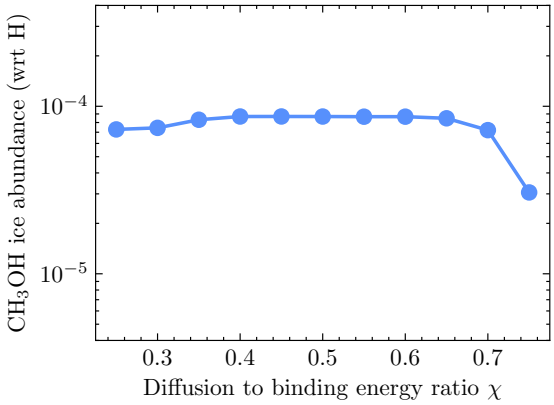

Figure 4: CH_3OH ice abundance for varying diffusion to binding energy ratio χ.

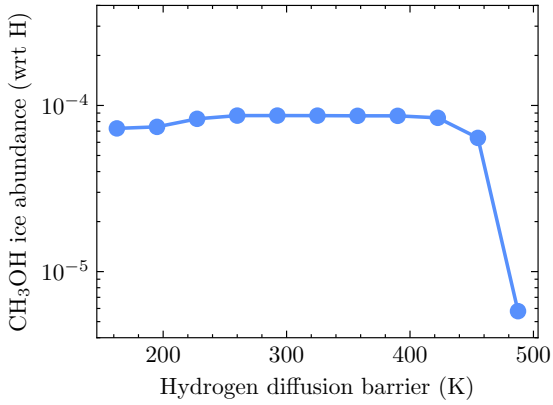

Figure 5: CH_3OH ice abundance for varying hydrogen diffusion barriers.

d. The two plots are basically identical, except at the drop-off at very high H diffusion barriers. This means that the most important parameter is the hydrogen diffusion barrier, and the other parameters that are varied in Fig. 4 are not so important. The fact that the drop-off is more severe in the H-only plot indicates that there is perhaps some diffusion-driven destruction reaction that is still relatively efficient (i.e. not limited by the slow H diffusion) in Fig. 5, but is slower in Fig. 4.

Code for solution:

```python
import numpy as np

import matplotlib.pyplot as plt
import uclchem
from uclchem.analysis import (
    get_production_and_destruction,
    rate_constants_to_dy_and_rates,
)
from uclchem.makerates.network import load_network_from_csv
from uclchem.plot import plot_rate_summary
from python_tools import convert_reaction_to_latex_string, convert_species_to_subscript

plt.style.use(["science", "font_latex", "color_petroff_6"])

if __name__ == "__main__":
    # set a parameter dictionary for model
    params = {
        "endAtFinalDensity": False,
        "freefall": False,
        "writeStep": 1,
        "initialDens": 1e5,
        "initialTemp": 10.0,
        "finalTime": 1e5,
        "zeta": 1,
        "radfield": 1,
    }

    # a
    physics_df, chemistry_df, rates_df, _, _, _ = uclchem.functional.cloud(
        param_dict=params, return_dataframe=True, return_rates=True
    )

    dy, rates_by_reaction_df = rate_constants_to_dy_and_rates(
        physics_df, chemistry_df, rates_df, network=load_network_from_csv()
    )

    spec = "#CH3OH"
    production, destruction = get_production_and_destruction(spec, rates_by_reaction_df)

    plot_rate_summary(
        production, destruction, -1, xlabel="Reaction rate (abundance wrt H s$^{-1}$)"
    )
    fig = plt.gcf()
    for axis in fig.axes:
        ticklabels: list[plt.Text] = axis.get_yaxis().get_ticklabels()
        axis.get_yaxis().set_ticklabels(
            [convert_reaction_to_latex_string(label.get_text()) for label in ticklabels]
        )
    plt.tight_layout()
    plt.savefig("rates_CH3OH_ice.pdf")
    plt.show()

```

```python
    # b
    step_size = 0.05
    diffusion_to_bind_min = 0.25
    diffusion_to_bind_max = 0.75
    num_steps = int((diffusion_to_bind_max - diffusion_to_bind_min) / (step_size)) + 1
    diffusion_to_bind_ratios, step = np.linspace(
        diffusion_to_bind_min,
        diffusion_to_bind_max,
        num=num_steps,
        endpoint=True,
        retstep=True,
    )
    assert step == step_size, (
        f"Expected step size {step_size}, but got step size {step}"
    )

    methanol_abundances = np.empty(len(diffusion_to_bind_ratios))
    for i, diffusion_to_bind_ratio in enumerate(diffusion_to_bind_ratios):
        print(f"Running chi = {diffusion_to_bind_ratio}")
        params["diffToBindRatio"] = float(diffusion_to_bind_ratio)

        physics_df, chemistry_df, _, _, _, _ = uclchem.functional.cloud(
            param_dict=params, return_dataframe=True
        )

        methanol_abundances[i] = (
            chemistry_df.iloc[-1]["#CH3OH"] + chemistry_df.iloc[-1]["@CH3OH"]
        )

    plt.figure()
    plt.plot(diffusion_to_bind_ratios, methanol_abundances, marker="o")
    plt.yscale("log")
    plt.ylabel(f"{convert_species_to_subscript('CH3OH')} ice abundance (wrt H)")
    plt.xlabel(r"Diffusion to binding energy ratio $\chi$")
    plt.ylim([4e-6, 4e-4])
    plt.tight_layout()
    plt.savefig("CH3OH_ice_against_chi.pdf")
    plt.show()

    # c
    step_size = 32.5
    h_diffusion_min = 162.5
    h_diffusion_max = 487.5
    num_steps = int((h_diffusion_max - h_diffusion_min) / (step_size)) + 1
    h_diffusion_barriers, step = np.linspace(
        h_diffusion_min,
        h_diffusion_max,
        num=num_steps,
        endpoint=True,
        retstep=True,
    )
    assert step == step_size, (
        f"Expected step size {step_size}, but got step size {step}"
    )

```

```python
    # Reset the diffusion to binding ratio chi to its default value
    params["diffToBindRatio"] = 0.5

    methanol_abundances = np.empty(len(h_diffusion_barriers))
    for i, h_diffusion_barrier in enumerate(h_diffusion_barriers):
        print(f"Running H diffusion barrier = {h_diffusion_barrier}")
        params["HdiffusionBarrier"] = float(h_diffusion_barrier)

        physics_df, chemistry_df, _, _, _, _ = uclchem.functional.cloud(
            param_dict=params, return_dataframe=True
        )

        methanol_abundances[i] = (
            chemistry_df.iloc[-1]["#CH3OH"] + chemistry_df.iloc[-1]["@CH3OH"]
        )

    plt.figure()
    plt.plot(h_diffusion_barriers, methanol_abundances, marker="o")
    plt.yscale("log")
    plt.ylabel(f"{convert_species_to_subscript('CH3OH')} ice abundance (wrt H)"
    )
    plt.xlabel(r"Hydrogen diffusion barrier (K)")
    plt.ylim([4e-6, 4e-4])
    plt.tight_layout()
    plt.savefig("CH3OH_ice_against_H_diffusion_barrier.pdf")
    plt.show()
```